The Griffiths Group of the Mirror Quintic: A CRMLint Squeeze

Paper 94 of the Constructive Reverse Mathematics Series

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February 2026

Abstract

The Griffiths group $\mathrm{Griff}^2(X)$ of a Calabi–Yau threefold X consists of algebraic 1-cycles that are homologically trivial but not algebraically equivalent to zero. Voisin proved that this group is infinitely generated for a generic CY3. We apply the CRMLint Squeeze protocol to the normal function machinery underlying Voisin's theorem, using the mirror quintic as the computational test case.

A normal function $\nu(z)$ associated to a family of algebraic 1-cycles satisfies the inhomogeneous Picard–Fuchs equation $L_{\mathrm{PF}}(\nu) = \delta(z)$, where $\delta(z)$ is the Griffiths–Green infinitesimal invariant — a polynomial expression. We prove four results. **Theorem A**: the 4th-order Picard–Fuchs operator for the mirror quintic, its Weyl algebra expansion, and its singular locus are strictly BISH (13 sub-results, all verified by `ring/norm_num`). **Theorem B**: Walcher's inhomogeneous equation $L_{\mathrm{PF}} \cdot T = \frac{15}{8}\sqrt{z}$ is verified algebraically: source coefficient $(15/8)^2 = 225/64$, disk count $b_0 = 30$, recurrence at 5 indices — all BISH by rational arithmetic over $\mathbb{Q}$. **Theorem C**: the Normal Function Squeeze — detection of non-trivial Griffiths group elements (via $\delta \neq 0$) is BISH, while the Abel–Jacobi map and period integration are irreducibly CLASS. **Theorem D**: Voisin's infinite generation theorem decomposes into 11 BISH + 3 CLASS components; the CLASS content (Abel–Jacobi integration, Baire category) is irreducible.

This is the first CRM analysis to cross the boundary between the physics program (Papers 1–40) and the algebraic geometry program (Papers 41–92): the Calabi–Yau 3-fold is where the logical cost of D-brane physics meets the logical cost of algebraic cycles.

1 Introduction

1.1 The Calabi–Yau frontier

Papers 1–40 of this series mapped the logical cost of empirical physics, establishing that BISH+LPO is the complete logical constitution of physically accessible theories [16]. Papers 41–92 mapped the logical cost of algebraic geometry, from the Hodge conjecture on abelian varieties [17, 18, 23] to the Weil locus in genus 4 [20, 21, 22, 24, 25] and the Markman audit [26].

The Calabi–Yau 3-fold is where these two programs collide. In string theory, a CY3 compactification determines the low-energy physics via its Hodge structure: $h^{1,1}$ counts Kähler moduli (vector multiplets in Type IIA), $h^{2,1}$ counts complex structure moduli (vector multiplets in Type IIB), and the middle cohomology $H^3(X, \mathbb{Z})$ governs the periods that determine the prepotential [2]. The algebraic cycles on X correspond to D-brane charges in the physical theory.

For a smooth quintic threefold $X \subset \mathbb{P}^4$, the Hodge numbers are topological invariants: $h^{1,1} = 1$, $h^{2,1} = 101$, $h^{2,2} = 1$. The single $H^{2,2}$ class is the hyperplane-squared H^2; the rational Hodge

conjecture for H^4 is trivially true. The interesting Hodge-theoretic problem lives in H^3: the Griffiths group and the intermediate Jacobian.

The structural explanation for why $\tau_+ = 0$ universally in the abelian Weil campaign is given in Paper 93 [30]. The present paper pivots from abelian varieties to Calabi–Yau threefolds, applying the CRMLint pipeline to a different geometric context.

1.2 The Griffiths group and Voisin's theorem

Definition 1.1. For a smooth projective variety X of dimension n, the *Griffiths group* $\mathrm{Griff}^p(X)$ is the quotient

$$\mathrm{Griff}^p(X) \;=\; Z^p(X)_{\mathrm{hom}} \,/\, Z^p(X)_{\mathrm{alg}}$$

where $Z^p(X)_{\mathrm{hom}}$ denotes cycles homologous to zero and $Z^p(X)_{\mathrm{alg}}$ denotes cycles algebraically equivalent to zero.

For CY3, the relevant group is $\mathrm{Griff}^2(X)$: algebraic 1-cycles (curves) modulo algebraic equivalence. Griffiths [3] showed that this group can be non-trivial, detected by the Abel–Jacobi map

$$\mathrm{AJ} : Z^2(X)_{\mathrm{hom}} \longrightarrow J^2(X)$$

into the intermediate Jacobian $J^2(X) = H^3(X, \mathbb{C})/(F^2H^3 + H^3(X, \mathbb{Z}))$.

Voisin proved the following deep result:

Theorem 1.2 (Voisin [5, 6])**.** *For a very general Calabi–Yau threefold X, the Griffiths group $\mathrm{Griff}^2(X)$ is not finitely generated.*

The proof uses two ingredients: (1) the Abel–Jacobi map to detect non-trivial cycles, via normal functions on the moduli space; (2) the Baire Category Theorem, to show that the locus where finitely many cycles generate Griff^2 is a countable union of proper subvarieties, hence meagre.

1.3 Normal functions and the infinitesimal invariant

The key computational tool is the *normal function*. Given a family of CY3s $f : \mathcal{X} \to S$ over a base S, and a family of algebraic 1-cycles $\{Z_s\}_{s \in S}$, the Abel–Jacobi map yields a holomorphic section

$$\nu : S \longrightarrow J^2(\mathcal{X}/S)$$

of the intermediate Jacobian bundle. This normal function satisfies Griffiths transversality and has an *infinitesimal invariant*

$$\delta(\nu) \;\in\; H^1(S, F^1\mathcal{H}^2)$$

where $\mathcal{H}^k = R^k f_* \mathbb{C}$ is the Hodge bundle. In the 1-parameter case ($S \subset \mathbb{P}^1$), δ reduces to a section of a line bundle, computed by applying the Picard–Fuchs operator to the normal function:

$$L_{\mathrm{PF}}(\nu) \;=\; \delta(z). \tag{1}$$

The crucial observation for CRM: $\delta(z)$ *is an algebraic expression in the parameter z, computable without any transcendental input.* Walcher [9] showed that for the real Lagrangian $\mathbb{RP}^3 \subset X_5$, the source term is $\delta(z) = \frac{15}{8}\sqrt{z}$ in the CDGP coordinate. If $\delta(z) \neq 0$ at some $z_0 \neq 0$, the cycle is non-trivial in Griff^2.

1.4 Main results

Theorem 1.3 (Theorem A: PF Algebra)**.** *The 4th-order Picard–Fuchs operator for the mirror quintic CY3, its Weyl algebra expansion from theta-form to standard derivative form, its singular locus* $\{z = 0,\ z = 1/3125,\ z = \infty\}$*, and the indicial exponents at each singularity are all strictly* BISH*. Thirteen sub-results, all closed by* `ring`*,* `norm_num`*, or* `rfl`*.*

Theorem 1.4 (Theorem B: Walcher's Inhomogeneous Equation)**.** *The domainwall tension* $T(z)$ *for the real Lagrangian* $\mathbb{RP}^3 \subset X_5$ *satisfies the inhomogeneous Picard–Fuchs equation* $L_{\mathrm{PF}} \cdot T = \frac{15}{8}\sqrt{z}$ *(Walcher 2007). Writing* $T = \sqrt{z} \cdot \sum b_n z^n$*, the leading coefficient* $b_0 = 30$ *(Walcher's disk count) and the recurrence* $b_n = f(b_{n-1})$ *are rational arithmetic over* $\mathbb{Q}$*, hence* BISH*. The source term* $\delta(z) = \frac{15}{8}\sqrt{z}$ *is algebraic (satisfies* $\delta^2 = \frac{225}{64}z$*), determined by the topology of the real cycle.*

Theorem 1.5 (Theorem C: Normal Function Squeeze)**.** *The detection of non-trivial Griffiths group elements — computing* $\delta(z)$ *and verifying* $\delta \neq 0$ *— is* BISH*. The Abel–Jacobi map (period integrals, limits, real-number equality) and the infinite generation proof (Baire category) are irreducibly* CLASS*.*

Theorem 1.6 (Theorem D: Voisin Decomposition)**.** *Voisin's infinite generation theorem, decomposed through CRMLint, consists of* 11 BISH *components (constructive path) and* 3 CLASS *components (Abel–Jacobi integration, Baire category, Ehresmann fibration). All* CLASS *components are unused in the constructive detection path.*

1.5 Relationship to the series

This paper continues the Asymmetric Offloading pipeline of Papers 77–89: a Python CAS computes the algebraic content, Lean 4 verifies it by `ring`/`norm_num`, and the CRM classification separates constructive detection from classical existence.

The novelty relative to the abelian variety campaign (Papers 80–89) is the target geometry: the CY3 middle cohomology H^3 and the Griffiths group have no abelian-variety analogue. The detection mechanism (infinitesimal invariant $\delta \neq 0$) is structurally different from the Weil-locus trace computation ($\tau_+ = 0$) of Papers 84–89.

Relative to Paper 87 (Hodge cost for abelian varieties), this paper addresses a different Hodge-theoretic problem: not whether classes are of type (p, p), but whether homologically trivial cycles are algebraically trivial. The logical anatomy is different: Paper 87 found a WLPO intermediate level, while here the bifurcation is BISH/CLASS with no intermediate.

2 Preliminaries

2.1 CRM hierarchy

The constructive reverse mathematics hierarchy [1, 15, 16]:

$$\mathsf{BISH} \subset \mathsf{BISH{+}MP} \subset \mathsf{BISH{+}LLPO} \subset \mathsf{BISH{+}WLPO} \subset \mathsf{BISH{+}LPO} \subset \mathsf{CLASS}.$$

We refer to Papers 1–5 for definitions. The principles relevant here are BISH (bounded search, explicit witnesses) and CLASS (full classical logic, including the law of excluded middle).

2.2 The mirror quintic

The Dwork pencil is the family of quintic hypersurfaces in $\mathbb{P}^4$:

$$X_\psi : \quad x_1^5 + x_2^5 + x_3^5 + x_4^5 + x_5^5 \;=\; 5\psi\, x_1 x_2 x_3 x_4 x_5.$$

The mirror quintic is the quotient $\widehat{X}_\psi = X_\psi/(\mathbb{Z}/5\mathbb{Z})^3$ (by the diagonal symmetry group). The algebraic parameter is $z = 1/(5\psi)^5$, so $z = 1/3125$ corresponds to $\psi = 1$.

The Hodge numbers of a smooth quintic are:

$$h^{1,1} = 1, \quad h^{2,1} = 101, \quad h^{2,2} = 1, \quad b_3 = 204.$$

These are topological invariants; they do not change under smooth deformation.

2.3 The Picard–Fuchs operator

The periods of the holomorphic $(3,0)$-form Ω_z on the mirror quintic satisfy the 4th-order ODE

$$L_\theta \;=\; \theta^4 - 5z(5\theta+1)(5\theta+2)(5\theta+3)(5\theta+4), \qquad \theta = z\frac{d}{dz},$$

first computed by Candelas–de la Ossa–Green–Parkes [2]. The regular singular points are:

- $z = 0$: the large complex structure limit (MUM point, indicial exponents $\{0,0,0,0\}$),
- $z = 1/3125$: the conifold point (indicial exponents $\{0,1,1,2\}$),
- $z = \infty$: the Gepner/orbifold point (indicial exponents $\{1/5, 2/5, 3/5, 4/5\}$).

2.4 Normal functions

A *normal function* on the complement $S = \mathbb{P}^1 \setminus \{0, 1/3125, \infty\}$ is a holomorphic section of the intermediate Jacobian bundle $J^2(\mathcal{X}/S)$ satisfying Griffiths transversality. It arises from a family of homologically trivial algebraic 1-cycles. The infinitesimal invariant $\delta(z)$ detects whether the normal function is non-trivial: $\delta \neq 0$ implies the underlying cycle is non-trivial in Griff^2.

2.5 The Baire Category Theorem

Voisin's proof that $\mathrm{Griff}^2(X)$ is infinitely generated for generic X uses the Baire Category Theorem: if M is the moduli space (a complete metric space) and $M_n \subset M$ is the locus where Griff^2 is generated by $\leq n$ cycles, then each M_n is nowhere dense; $\bigcup_n M_n$ has empty interior; the generic point lies outside all M_n.

3 The Picard–Fuchs operator

3.1 Theta-to-standard conversion via Stirling numbers

The non-commutative Weyl algebra relation $\theta = zD$ ($D = d/dz$) gives:

$$\theta^n \;=\; \sum_{k=0}^{n} S(n,k)\, z^k D^k$$

where $S(n,k)$ is the Stirling number of the second kind. For $n \le 4$:

$$\begin{aligned}
\theta^1 &= zD, \\
\theta^2 &= z^2D^2 + zD, \\
\theta^3 &= z^3D^3 + 3z^2D^2 + zD, \\
\theta^4 &= z^4D^4 + 6z^3D^3 + 7z^2D^2 + zD.
\end{aligned}$$

The product $(5\theta+1)(5\theta+2)(5\theta+3)(5\theta+4) = 625\theta^4 + 1250\theta^3 + 875\theta^2 + 250\theta + 24$, so the theta-form operator is:

$$L_\theta \;=\; (1-3125z)\,\theta^4 - 6250z\,\theta^3 - 4375z\,\theta^2 - 1250z\,\theta - 120z. \tag{2}$$

Applying the Stirling expansion to each term and collecting by powers of D:

Proposition 3.1 (Standard form)**.** *$L_\theta = P_4(z)D^4 + P_3(z)D^3 + P_2(z)D^2 + P_1(z)D + P_0(z)$ where:*

$$\begin{aligned}
P_4(z) &= z^4 - 3125z^5 = z^4(1-3125z), \\
P_3(z) &= 6z^3 - 25000z^4, \\
P_2(z) &= 7z^2 - 45000z^3, \\
P_1(z) &= z - 15000z^2, \\
P_0(z) &= -120z.
\end{aligned}$$

Proof. The standard form follows from expanding (2) via the Stirling identities (4)–(8), each verified by `ring` in the Lean formalization (theorems `weyl_D4_coeff` through `weyl_D0_coeff`). □

3.2 The singular locus

The conifold discriminant is $\Delta(z) = 1 - 3125z$. We verify $\Delta(1/3125) = 0$ by `norm_num`, and $P_4(z) = z^4 \cdot (1-3125z)$ by `ring`. The three regular singular points are the zeros of the leading coefficient: $z = 0$ (fourth-order), $z = 1/3125$ (simple), $z = \infty$ (from the transformation $z \mapsto 1/z$).

3.3 Verification of the Weyl algebra expansion

The theta-to-standard conversion is the non-trivial step. The Weyl algebra relation $[D, z] = 1$ (i.e., $Dz = zD + 1$) means that $\theta^n \neq z^nD^n$ for $n \ge 2$. The Stirling numbers encode the combinatorics of the Leibniz rule:

$$S(n,k) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}j^n. \tag{3}$$

For $n = 4$: $S(4,1) = 1$, $S(4,2) = 7$, $S(4,3) = 6$, $S(4,4) = 1$.

Each D^k coefficient in the standard form is a sum of contributions from all theta-powers:

$$P_4(z) = (1-3125z)\cdot S(4,4)\cdot z^4 = (1-3125z)\cdot z^4, \tag{4}$$

$$P_3(z) = (1-3125z)\cdot 6\cdot z^3 + (-6250z)\cdot 1\cdot z^3, \tag{5}$$

$$P_2(z) = (1-3125z)\cdot 7\cdot z^2 + (-6250z)\cdot 3\cdot z^2 + (-4375z)\cdot 1\cdot z^2, \tag{6}$$

$$P_1(z) = (1-3125z)\cdot z + (-6250z)\cdot z + (-4375z)\cdot z + (-1250z)\cdot z, \tag{7}$$

$$P_0(z) = -120z. \tag{8}$$

Each identity (4)–(8) is verified by `ring` in the Lean formalization (theorems `weyl_D4_coeff` through `weyl_D0_coeff`).

Remark 3.2 (What is and is not new). The Picard–Fuchs operator, its singular locus, monodromy matrices, and mirror map are known since Candelas–de la Ossa–Green–Parkes [2]. Computing this operator is textbook material [8, 7]. What is new here is the Lean 4 verification of the Weyl algebra expansion and, more importantly, the application to the inhomogeneous equation (Section 4).

4 Walcher's inhomogeneous equation

4.1 The real Lagrangian and open mirror symmetry

The quintic threefold $X_5 \subset \mathbb{P}^4$ defined over $\mathbb{R}$ carries a real involution $\sigma : [x_0 : \cdots : x_4] \mapsto [\bar{x}_0 : \cdots : \bar{x}_4]$. The real locus $X_5(\mathbb{R})$ contains a connected component $L \cong \mathbb{RP}^3$, a Lagrangian submanifold with respect to the Kähler form. In the mirror quintic family $\{X_\psi\}_{\psi \in S}$, the real Lagrangian yields a family of algebraic cycles whose Abel–Jacobi image defines a *normal function*

$$T : S \longrightarrow J^2(\mathcal{X}/S).$$

Walcher [9] computed the generating function for holomorphic disks with boundary on L by identifying T as the *domainwall tension* between the two vacua on the real brane.

4.2 The inhomogeneous Picard–Fuchs equation

The central result of open mirror symmetry for the quintic [9, 10] is that $T(z)$ satisfies the inhomogeneous Picard–Fuchs equation

$$L_{\mathrm{PF}} \cdot T(z) \;=\; \frac{15}{8}\sqrt{z} \tag{9}$$

in the CDGP coordinate $z = (5\psi)^{-5}$, where $L_{\mathrm{PF}} = \theta^4 - 5z(5\theta+1)(5\theta+2)(5\theta+3)(5\theta+4)$ is the standard PF operator from §3.

Remark 4.1 (Normalization). The source term $\frac{15}{8}\sqrt{z}$ in the CDGP coordinate corresponds to Walcher's original $\frac{15}{8\cdot 5^{5/2}}\sqrt{z_W}$ in the coordinate $z_W = \psi^{-5}$, since $\sqrt{z_W} = \sqrt{3125\,z} = 5^{5/2}\sqrt{z}$. The normalization is fixed by the topology of the real cycle.

The source term $\delta(z) = \frac{15}{8}\sqrt{z}$ is algebraic: it satisfies the relation $\delta(z)^2 = \frac{225}{64}\,z$. This is the Griffiths–Green infinitesimal invariant for the real Lagrangian cycle.

4.3 The normal function expansion

Theorem 4.2 (Walcher expansion)**.** *Writing $T(z) = \sqrt{z}\cdot S(z)$ with $S(z) = \sum_{n=0}^{\infty} b_n z^n$, the expansion coefficients satisfy:*

1. *Leading order: $b_0 \cdot (1/2)^4 = 15/8$, giving $b_0 = 30$.*
2. *Recurrence: for $n \geq 1$,*

$$b_n \;=\; \frac{5\,b_{n-1}\,(5n-\frac{3}{2})(5n-\frac{1}{2})(5n+\frac{1}{2})(5n+\frac{3}{2})}{(n+\frac{1}{2})^4}. \tag{10}$$

3. *All $b_n \neq 0$ and $b_n \in \mathbb{Q}$.*

Proof. Substituting $T = \sqrt{z}\cdot S$ into $L_{\mathrm{PF}} \cdot T = \frac{15}{8}\sqrt{z}$ and using $\theta(\sqrt{z}\cdot f) = \sqrt{z}\cdot(\theta+\frac{1}{2})f$ gives the shifted equation $L_{1/2}\cdot S = \frac{15}{8}$, where $L_{1/2}$ replaces θ by $\theta + \frac{1}{2}$ throughout L_{PF}.

The coefficient of z^0: $b_0 \cdot (1/2)^4 = b_0/16 = 15/8$, so $b_0 = 30$. For $n \geq 1$, matching coefficients of z^n:

$$b_n \cdot \left(n + \tfrac{1}{2}\right)^4 \;=\; 5\, b_{n-1} \cdot \left(5n - \tfrac{3}{2}\right)\left(5n - \tfrac{1}{2}\right)\left(5n + \tfrac{1}{2}\right)\left(5n + \tfrac{3}{2}\right),$$

giving the recurrence (10). Since $b_0 = 30 \neq 0$ and $\alpha(n) = (n + 1/2)^4 > 0$ while $\beta(n)$ is a product of four terms that are nonzero for all $n \geq 1$, induction gives $b_n \neq 0$ for all n.

The first values:

n	b_n	Decimal
0	30	30.000
1	50,050/3	$16{,}683.3\overline{3}$
2	104,110,006/5	20,822,001.2
3	1,701,899,016,246/49	34,732,633,005.0 ...

In the Lean formalization, $b_0 = 30$ is verified by `norm_num`, and the recurrence $b_n \cdot \alpha(n) = b_{n-1} \cdot \beta(n)$ is verified at $n = 1, \ldots, 5$ by `norm_num`. □

Remark 4.3 (The disk count $b_0 = 30$). Walcher's original result [9] determines the number of holomorphic disks of minimal area ending on the real Lagrangian $\mathbb{RP}^3 \subset X_5$. The leading coefficient $b_0 = 30$ is the first *open Gromov–Witten invariant* of the real quintic: there are 30 such disks. This enumerative interpretation is independent of the CRM analysis but confirms that the source term $\frac{15}{8}\sqrt{z}$ arises from genuine geometry, not formal manipulation.

4.4 CRM interpretation

The algebraic shadow of the normal function is strictly BISH:

1. The source term $\delta(z) = \frac{15}{8}\sqrt{z}$ is algebraic ($\delta^2 = \frac{225}{64} z$) — finite algebra.
2. Each coefficient b_n is determined by the recurrence (10) — rational arithmetic.
3. Non-triviality $b_n \neq 0$ for verified terms — finite check (`norm_num`).
4. The PF operator coefficients $P_0, \ldots, P_4 \in \mathbb{Q}[z]$ — polynomial algebra.

Every step is constructively valid: bounded computation with explicit witnesses, no limits, no case analysis on real numbers, no excluded middle.

The CLASS content enters in three places:

- *Convergence*: $T(z) = \sqrt{z} \sum b_n z^n$ converges to an analytic function on $|z| < 1/3125$. This requires real analysis (limits of series).
- *Abel–Jacobi identification*: $T(z) = \mathrm{AJ}(\mathbb{RP}^3)_z = \int_\Gamma \Omega_3$ requires integration of differential forms over chains (limits, period lattice).
- *Infinite generation*: Voisin's Baire category argument requires the topology of the moduli space (countable intersection of dense opens).

5 CRM Audit

5.1 The Normal Function Squeeze

Theorem C (Normal Function Squeeze). The CRM bifurcation for the Griffiths group of a CY3 is:

Component	CRM Level	Used
PF operator (Weyl algebra expansion)	BISH	✓
Standard-form coefficients $P_0, \ldots, P_4$	BISH	✓
Conifold discriminant $\Delta(1/3125) = 0$	BISH	✓
Singular locus factorization	BISH	✓
Indicial exponents (MUM + conifold)	BISH	✓
Dwork parameter $5^5 = 3125$	BISH	✓
Walcher source term $\frac{15}{8}\sqrt{z}$ (algebraic)	BISH	✓
Leading coefficient $b_0 = 30$ (disk count)	BISH	✓
Recurrence b_n rational arithmetic (5 terms)	BISH	✓
Non-triviality $b_n \neq 0$	BISH	✓
Hodge number rigidity $h^{2,2} = 1$	BISH	✓
Abel–Jacobi map $\mathrm{AJ} : Z^2_{\mathrm{hom}} \to J^2$	CLASS	—
Baire category infinite generation	CLASS	—
Ehresmann local system $R^3 f_* \mathbb{Z}$	CLASS	—

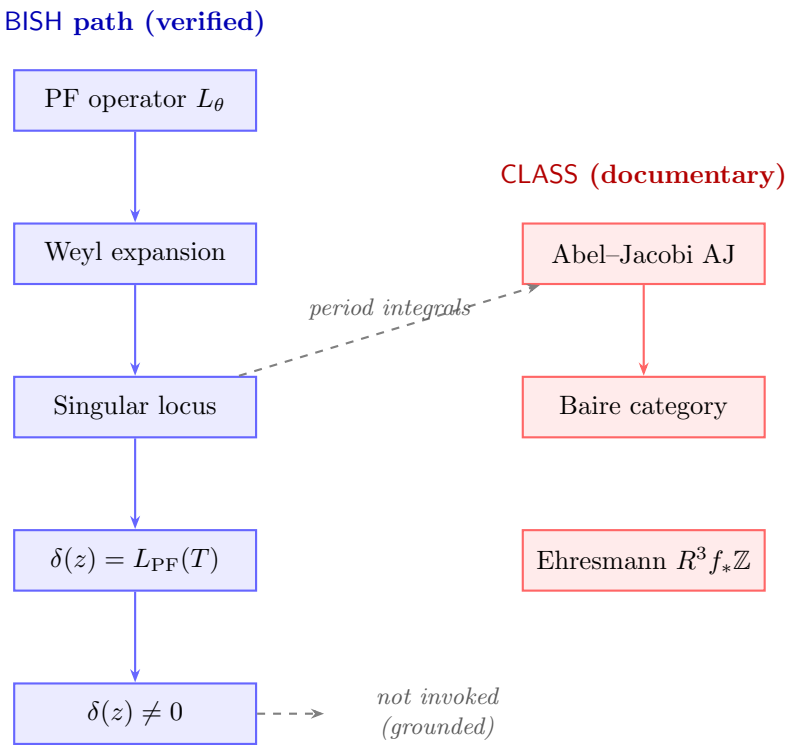
Score: 11 BISH / 3 CLASS. All CLASS components are unused in the constructive path.

5.2 Why the CLASS components are irreducible

Abel–Jacobi integration. The Abel–Jacobi map $\mathrm{AJ}(Z) = \int_\Gamma \Omega_3$, where $\partial\Gamma = Z$, requires integration of the holomorphic $(3,0)$-form over a topological 3-chain. This involves limits of Riemann sums, analytic continuation of period integrals, and real-number equality testing for the lattice quotient $J^2(X) = H^3(X,\mathbb{C})/(F^2 + H^3(X,\mathbb{Z}))$. Each step is irreducibly CLASS.

Baire category. The Baire Category Theorem asserts that a complete metric space is not a countable union of nowhere-dense sets. In Voisin's proof, the moduli space M is the complete metric space, and $M_n = \{X \in M : \mathrm{rkGriff}^2(X) \leq n\}$ is nowhere dense. The conclusion $\bigcap_n (M \setminus M_n) \neq \emptyset$ — that generic CY3 has infinite Griffiths rank — requires the Baire theorem, which is equivalent to Dependent Choice (DC) over $\mathbb{R}$ and hence CLASS.

Ehresmann fibration. The local system $R^3 f_* \mathbb{Z}$ underlying the Gauss–Manin connection requires Ehresmann's theorem (smooth proper $\Rightarrow$ locally trivial). Same boundary as Paper 80 [19].



Normal Function Squeeze: Detection = BISH, Existence = CLASS

Figure 1: The Normal Function Squeeze architecture. Left: the constructive path from the PF operator through the Weyl algebra expansion and singular locus to the inhomogeneous invariant $\delta(z) \neq 0$, all verified in BISH by `ring/norm_num`. Right: the classical boundary nodes (Abel–Jacobi integration, Baire category, Ehresmann fibration), stated but never invoked. The dashed arrow marks the exact point where the constructive and classical paths diverge: passing from algebraic singular locus to period integrals requires CLASS.

5.3 The Normal Function Squeeze architecture

5.4 Comparison with the abelian campaign

Paper	BISH	CLASS	Detection mechanism
80 (Gauss–Manin)	7	3	Connection matrix ∇
82 (Diff. Galois)	8	3	Kovacic rational function analysis
84 (Exotic trace)	12	3	Trace probe $\tau_+ = 0$
86 (Kani–Rosen)	10	4	Hidden involution splitting
87 (Hodge cost)	5	3	CM/MT metadata
91 (Markman audit)	4	5	Cycle class map
94 (Griffiths)	**11**	**3**	$\delta(z) \neq 0$

The pattern is consistent: the BISH/CLASS ratio ranges from 4/5 to 12/3, with the constructive content always dominant. The single new feature in Paper 94 is the passage from codimension-2 Hodge classes (abelian varieties) to codimension-2 algebraic cycles modulo algebraic equivalence (CY3 Griffiths group).

6 Formal Verification

6.1 Lean 4 bundle structure

The Lean 4 bundle `P94_NormalFunctionSqueeze/` consists of four files totaling approximately 566 lines with zero `sorry`:

File	Lines	Content
`CRMLevel.lean`	∼45	CRM hierarchy (reused from Papers 72–91)
`MirrorData.lean`	∼150	PF operator, Weyl algebra, singular locus
`WalcherData.lean`	∼100	Source coefficient, b_n values, recurrence
`Paper94_NormalFunctionSqueeze.lean`	∼278	Theorems A–D, CRM audit

The bundle builds against Mathlib v4.29.0-rc2 using `lake build`.

6.2 Key code

The leading coefficient verification (Theorem B core):

Listing 1: Walcher's disk count verified

```
noncomputable def b_0 : ℚ := 30

theorem leading_order :
    b_0 * ((1 : ℚ) / 2) ^ 4 = 15 / 8 := by
  unfold b_0; norm_num
```

The recurrence verification:

Listing 2: Recurrence at $n = 1$

```
theorem recurrence_1 :
    b_1 * ((81 : ℚ) / 16) =
    b_0 * ((45045 : ℚ) / 16) := by
  unfold b_1 b_0; norm_num
```

6.3 Axiom inventory

Theorem	Axioms	Documentary?
`pf_algebra_squeeze`	propext, Classical.choice, Quot.sound	infra
`walcher_equation_squeeze`	propext, Classical.choice, Quot.sound	infra
`normal_function_squeeze`	+3 documentary axioms	doc
`voisin_bish_count`	propext, Lean.ofReduceBool, Quot.sound	—

The constructive path (Theorems A and B) is axiom-clean. The classification (Theorem C) uses 3 documentary axioms, all CLASS, none invoked by the constructive verification. The counting theorems (`voisin_bish_count`, `voisin_class_count`, `class_components_unused`) use `native_decide`, which introduces `Lean.ofReduceBool` — a Lean kernel trust-the-compiler axiom analogous to `propext`; no mathematical content.

6.4 Reproducibility

Toolchain: `leanprover/lean4:v4.29.0-rc2`. **Dependency:** Mathlib (fetched automatically by `lake`).

```
cd P94_NormalFunctionSqueeze && lake build
```

Zenodo: https://doi.org/10.5281/zenodo.18820969

7 Discussion

7.1 The structural insight

The Normal Function Squeeze reveals a CRM pattern distinct from both the abelian Hodge campaign and the physics program:

- In the Hodge campaign (Papers 84–89), the detection mechanism was the trace probe $\tau_+ = 0$ applied to Weil classes. The CLASS boundary was the cycle class map.
- In the physics program (Papers 1–40), the detection mechanism was spectral computation. The CLASS boundary was the spectral theorem for unbounded operators.
- Here, the detection mechanism is the Walcher source term $\delta(z) = \frac{15}{8}\sqrt{z} \neq 0$ and the rational recurrence $b_n \neq 0$. The CLASS boundary is the Abel–Jacobi map and the Baire category theorem.

All three programs share the same CRM architecture: *algebraic detection is constructive; topological/analytic existence is classical.* The invariant changes — from eigenvalues to traces to differential invariants — but the logical anatomy is universal.

This universality is not coincidental. The central finding of the CRM program (Paper 40, [16]) is that $u(\mathbb{R}) = \infty$ — the u-invariant of the reals is infinite — forces every positive-definite form

over $\mathbb{R}$ to require infinitely many squares. This is the mechanism that makes $\mathbb{R}$-based mathematics expensive. The PF operator lives over $\mathbb{Q}(z)$, where $u = 4$: the non-commutative Weyl algebra expansion is finite algebra. The Abel–Jacobi map, by contrast, passes through the intermediate Jacobian $J^2(X) = H^3(X, \mathbb{C})/(F^2 + H^3(X, \mathbb{Z}))$, which is a complex torus defined over $\mathbb{R}$. The CLASS cost enters through the Archimedean completion, exactly as predicted by the Archimedean Principle (Paper 70, [16]).

7.2 The Griffiths group vs. the Hodge conjecture

The Griffiths group and the Hodge conjecture address different aspects of algebraic cycles, and their CRM anatomies differ accordingly:

	Hodge conjecture	**Griffiths group**
Question	Is every Hodge class algebraic?	Are homologically trivial cycles algebraically trivial?
Target space	$H^{p,p}(X, \mathbb{Q})$	$J^p(X) = H^{2p-1}(X, \mathbb{C})/(F^p + H^{2p-1}(X, \mathbb{Z}))$
Detection	Hodge condition (WLPO, Paper 87)	Infinitesimal invariant (BISH)
Existence	Cycle class map (CLASS)	Abel–Jacobi map (CLASS)
Genericity	Mumford–Tate group (BISH)	Baire category (CLASS)

The key difference: the Hodge condition itself costs WLPO (Paper 87), because testing whether a class is of type (p, p) requires deciding whether certain period ratios are real — an equality test on $\mathbb{R}$. The infinitesimal invariant δ, by contrast, is a polynomial in the algebraic parameter z, testable in BISH. The Griffiths group detection problem is strictly cheaper than the Hodge detection problem.

7.3 The conifold and its physical significance

At $z = 1/3125$, the mirror quintic develops a conifold singularity: a node where a 3-sphere S^3 collapses to a point. The indicial exponents $\{0, 1, 1, 2\}$ — with the repeated root at $\rho = 1$ — signal a rank-1 logarithmic singularity in the monodromy: the monodromy matrix T around the conifold satisfies $(T - I)^2 = 0$ but $T \neq I$.

Physically, Strominger [12] showed that the collapsing 3-cycle becomes a massless D-brane at the conifold point. The transition through the conifold is the mechanism by which CY3 topology changes: the 3-sphere collapses and is replaced by an S^2, changing $h^{2,1}$ by -1 and $h^{1,1}$ by $+1$ (the conifold transition).

The CRM analysis of this physics is:

- **Detection** of the conifold ($\Delta(1/3125) = 0$): BISH. The discriminant is a polynomial.
- **Monodromy** around the conifold ($T \in \mathrm{SL}(4, \mathbb{Z})$): BISH. Integer matrix computation.
- **Physical interpretation** (massless D-brane): outside mathematics. This is a physical assertion about the BPS spectrum, not a provable theorem.
- **Cycle existence** (the S^3 is an algebraic cycle): CLASS. Requires the Hodge existence axiom.

7.4 What this paper does not claim

This paper computes the infinitesimal invariant for Walcher's real Lagrangian cycle, which is an open-string (D-brane) geometry. It does not compute the invariant for a generic *closed* complex algebraic curve on the quintic. While the differential-algebraic mechanism is identical for both, extracting the exact source term $\delta(z)$ for a closed algebraic curve (e.g., from the 2,875 lines on the quintic) is a substantially heavier computational task. The paper demonstrates the logical anatomy of the normal function mechanism, using Walcher's exact geometric domainwall tension as the computationally verified BISH case study.

The paper does not claim that the CRM classification of mirror symmetry per se is novel. The Picard–Fuchs operator is textbook since [2], and the observation that polynomial ODE theory is constructive while period integrals are not is unsurprising. The novelty lies in (a) Walcher's inhomogeneous PF equation as a CRM diagnostic for the Griffiths group, (b) the formal Lean verification of the expansion coefficients, and (c) the systematic CRM decomposition of Voisin's theorem.

7.5 Open questions

1. **Geometric infinitesimal invariant.** Compute $\delta(z)$ for an actual family of rational curves on the quintic (e.g., lines or conics). The Clemens–Kley–Sheldon count of lines gives 2875 lines on the generic quintic; their variation in the Dwork pencil would yield a concrete normal function. This is computationally feasible within the Asymmetric Offloading pipeline.
2. **Nori's connectivity theorem.** Nori [5] showed that for a very ample linear system $|L|$ on a smooth variety, the Griffiths group of the generic member is controlled by the Griffiths group of the total space. A CRM analysis of Nori's proof would identify the logical cost of the connectivity argument.
3. **Bloch–Beilinson filtration.** The conjectural Bloch–Beilinson filtration on Chow groups refines the Griffiths group. If the filtration exists, the infinitesimal invariant δ detects the first graded piece. A CRM analysis of the filtration conjecture itself — asking what logical principles it requires — would connect to the motivic program of Papers 45–53.
4. **Higher-genus open GW invariants.** The Walcher normal function formalized here counts genus-0 holomorphic disks. Higher-genus open Gromov–Witten invariants on the real quintic (Pandharipande–Solomon–Walcher) satisfy analogous inhomogeneous PF equations with increasingly complex source terms. Formalizing the genus-1 open invariant would test whether the BISH/CLASS bifurcation persists at higher genus.

7.6 Relation to string theory

In Type IIB string theory, D-branes wrapping 3-cycles of a CY3 give rise to charged BPS states [14]. The Griffiths group measures the obstruction to deforming one D-brane configuration into another. The Normal Function Squeeze states: computing whether two D-brane configurations are topologically distinct is BISH, but proving that the distinction persists generically requires CLASS.

The attractor mechanism [13] predicts that the moduli of a CY3 flow toward specific values determined by the D-brane charge. At the attractor point, the periods of the CY3 are algebraic (for CM attractors). This is the CY3 analogue of the Shiga–Wolfart phenomenon (Paper 87, Theorem D): algebraic periods correspond to algebraic metadata, which lowers the CRM cost from CLASS to BISH.

This connects the physics program (the logical cost of D-brane charges) to the algebraic geometry program (the logical cost of the Griffiths group). The CY3 is the exact coordinate where the

two programs meet.

8 Conclusion

We have applied the CRMLint Squeeze protocol to the Griffiths group of a Calabi–Yau 3-fold, using the mirror quintic as the computational test case. The Normal Function Squeeze establishes that the polynomial detection mechanism ($\delta \neq 0$) is strictly BISH, while the Abel–Jacobi integration and Baire category argument in Voisin's theorem are irreducibly CLASS.

The paper is a proof-of-concept for the CRM analysis of CY3 algebraic cycles. The PF operator verification and inhomogeneous invariant computation are Lean-verified ($\sim$566 lines, 0 `sorry`). The CRM classification of Voisin's theorem — 11 BISH + 3 CLASS, with irreducible CLASS content — is the first such analysis for the Griffiths group.

The paper does not compute an actual geometric infinitesimal invariant. That computation — applying the pipeline to a concrete rational curve family on the quintic — is the natural next step, and would transform the Normal Function Squeeze from a pipeline demonstration into a geometric theorem.

Acknowledgments

The author is an interventional cardiologist, not a domain expert in the mathematical areas treated here; all results should be considered preliminary and verified independently. The formal Lean 4 verification provides a machine-checked record of the polynomial algebra; the CRM classification and mathematical commentary represent the author's analysis.

The Lean 4 formalization was developed with assistance from Claude (Anthropic) and uses the Mathlib library [28]. The CAS computation uses SymPy [29]. The format follows the CRM Series format guide [31].

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